

.NET Self-Learning Questions

Session 2 – .NET Fundamentals

Study & Understand (Write Answers)

1. Write a comparison between .NET Framework, .NET Core, and .NET 8.
2. Write the differences between CLR and JVM.
3. Write Managed Code and Unmanaged Code with suitable examples.
4. Write the purpose of CLS and CTS in .NET.
5. Describe the role of JIT Compiler in the .NET execution lifecycle.

Implementation Questions

1. Create a Console Application and inspect the generated IL code using ILDASM.
 2. Identify IL instructions generated for:
 - Method Calls
 - Object Creation
 - Exception Handling
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Session 4 – Static Members & Inheritance

Study & Understand (Write Answers)

1. Differentiate between Static Constructor and Instance Constructor.
2. Write the restrictions of a Static Class.
3. Write Method Overriding and Method Hiding.
4. What are Sealed Methods? Write with examples.
5. Write Singleton Pattern and Template Method Pattern.

Implementation Questions

1. Create a program demonstrating Static Constructor and Instance Constructor.
 2. Implement Constructor Chaining using `base()` and `this()`.
 3. Create a class hierarchy demonstrating Method Overriding.
 4. Demonstrate Method Hiding using the `new` keyword.
 5. Implement Singleton Pattern.
 6. Implement Template Method Pattern.
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Session 7 – Generics & Collections

Study & Understand (Write Answers)

1. When should List be preferred over Dictionary?
2. Write the use cases of HashSet, Queue, and Stack.

3. Write the internal working of Dictionary with hashing and collision concepts.
4. Differentiate between IComparable and IComparer.

Implementation Questions

1. Implement a Shopping Cart System using appropriate collections.
 2. Implement generic sorting using IComparable.
 3. Implement generic sorting using IComparer.
 4. Create examples using List, Dictionary, Queue, Stack, and HashSet.
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Session 9 – Exception Handling & Events

Study & Understand (Write Answers)

1. Differentiate between `throw;` and `throw ex;`.
2. Write the purpose of `checked` and `unchecked`.
3. What are Inner Exceptions?
4. Write the Observer Pattern.
5. Compare Observer Pattern using Interfaces and Events.

Implementation Questions

1. Create a custom exception hierarchy:
 - PaymentException
 - InventoryException
 - ValidationException
 2. Demonstrate Multiple Catch Blocks.
 3. Demonstrate Inner Exceptions.
 4. Implement Observer Pattern using Interfaces.
 5. Implement Observer Pattern using Events and Delegates.
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Session 10 – LINQ & Extension Methods

Study & Understand (Write Answers)

1. Write Deferred Execution in LINQ.
2. What are Side Effects in LINQ?
3. Compare LINQ with Java Streams.
4. Write the purpose of `SelectMany()`, `GroupBy()`, and `ToDictionary()`.

Implementation Questions

1. Create a Product Inventory System using LINQ.
2. Filter products by price.

3. Group products by category.
 4. Order products using multiple fields.
 5. Calculate aggregate totals.
 6. Demonstrate `SelectMany()`, `GroupBy()`, and `ToDictionary()`.
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Session 11 – Reflection, Attributes & File I/O

Study & Understand (Write Answers)

1. Write Reflection in .NET.
2. Compare Reflection in .NET and Java.
3. Write JSON Serialization.
4. Write IDisposable Pattern and `using` Statement.
5. Write the purpose of HttpClient.

Implementation Questions

1. Implement an ORM SQL Generator using Reflection and Attributes.
 2. Implement a File-Based Logging System.
 3. Write a program to consume a REST API using HttpClient.
 4. Serialize and Deserialize objects using JSON.
 5. Demonstrate IDisposable and `using` statement.
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Sessions 13–22 – ASP.NET Core, EF Core, Web API & React

Study & Understand (Write Answers)

1. Compare ADO.NET and Entity Framework.
2. Write the advantages of EF Core.
3. Write the architecture of ASP.NET Core Web API.
4. Write the deployment process of a Web API on IIS.
5. Write React and ASP.NET Core Integration.

Implementation Questions

1. Implement CRUD Operations using EF Core for the food Ordering System
2. Implement a Web API for the food Ordering System
3. Deploy the Web API to IIS.
4. Implement a React Application integrated with the food Ordering System.